Cover Page

- **Concept Title: AI-Based Disaster Damage Assessment & Rescue Priority Mapping**
- **Theme: Disaster Management / AI for Social Impact**
- **Team Name: TriZen**
- **Team ID: HC|TN|02694**
- **State/UT: Tamil Nadu**

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Problem – Proposed Solution

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Core concept

An AI system analyzes satellite or drone images after a disaster to identify damaged buildings, blocked roads, and affected zones.

How it works

• Upload imagery → AI detects damage and road blockages → classify severity → create a color-coded rescue-priority map.

• High-priority zones are highlighted for rapid field verification and resource deployment.

What makes it innovative

• Combines damage detection, severity ranking, and map-based prioritization in one workflow.

• Helps responders focus first on locations likely to need urgent rescue or access clearance.

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Technical Approach

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Suggested technology stack

• Python • OpenCV • PyTorch/TensorFlow • YOLO or image-segmentation model

• QGIS/GeoPandas • Leaflet/Folium map • Streamlit or Flask dashboard

Implementation flow

Satellite/drone images → preprocessing → AI damage detection → severity classification → geospatial tagging → priority map/dashboard

AI usage

AI/ML is the core component for image analysis; human review is recommended before operational decisions.

GitHub/GitLab: [Add repository link] Demo video: [Add demo link]

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Impact Potential

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Target users

• Disaster response teams • Fire and rescue services • Local authorities • NGOs • Relief coordinators

Expected benefits

• Faster initial assessment of large affected areas
• Clearer visibility of damaged infrastructure and blocked routes
• Better prioritization of field inspections, rescue teams, and supplies
• Map-based information supports coordination across agencies

Social impact

Can help reduce assessment delays and support more informed response decisions. Outputs should be verified by trained responders.

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Feasibility and Risk Analysis

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Feasibility

- Prototype can be built using open-source computer-vision and mapping tools.
- Public satellite imagery or sample drone images can support initial testing.

Risks and challenges → Mitigation

- Cloud, smoke, low resolution → combine available image sources; flag low-confidence results.
- Limited/biased training data → test across different building types and disaster scenes.
- Network or imagery delays → allow offline map export and timestamp every image.
- False positives/negatives → show confidence scores and require human validation.

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Team

Team composition (edit to match your registered team)

- AI / Computer Vision: model selection, training, and evaluation
- Software / Dashboard: image upload, map interface, and results display
- GIS / Data: geospatial tagging, map layers, and data preparation
- Research / Presentation: validation, documentation, and stakeholder needs

Likely challenge

Getting reliable labeled disaster imagery. We will start with a small proof-of-concept dataset, evaluate errors, and improve the model iteratively.

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References

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Suggested references to consult and cite in the final submission

• NASA Earth Observatory — satellite imagery and disaster observations: https://earthobservatory.nasa.gov/
• Copernicus Emergency Management Service: https://emergency.copernicus.eu/
• UNOSAT — satellite analysis for humanitarian response: https://unosat.org/
• QGIS Documentation — geospatial mapping tools: https://docs.qgis.org/
• Ultralytics YOLO Documentation — object detection: https://docs.ultralytics.com/

Note: Add any datasets, model papers, or imagery sources actually used in your prototype.